Emerging broadband access technologies

D E A Clarke, G Lampard, R P Merrett, D W Smith, J F Wakeling, T J Whitley and G Young

This paper briefly outlines the wide variety of emerging technologies that are set to challenge the traditional copper pair access network for new services. Features such as capacity and geographical coverage of the alternatives are discussed.

1. Introduction

The expression 'the local loop' encompasses a 100-year tradition of a copper pair access network supporting analogue voice transmission and loop-disconnect signalling. Although the term 'local loop' remains, what is now happening in access networks is anything but traditional. Today, for the larger business customer, that 'loop' may well be high-bandwidth single-mode optical fibre. Even the small and medium scale user may be connected by alternative access technologies, such as terrestrial radio, cable or satellite. Each of these new access technologies has different performance, geographical and economic scaling attributes. Satellite technologies may appeal where a new entrant has a low penetration over a wide geographic area but may not scale so well in a dense urban location. Emerging access technology is leading to increasingly complex investment decisions for both long-established operators and new entrants. This is also making it necessary to continuously reconsider how much more can be squeezed out of existing copper plant before deciding to invest in future-proof optical fibre.

This paper provides a brief overview of the various access technologies that have emerged in the past few years and offers a perspective on their potential use within the ever more complex telecommunications market-place.

2. Emerging broadband access technologies

2.1 Networks in the sky

Traditional satellite communications rely on so-called 'geostationary' satellites (Fig 1). This term describes a satellite placed in an orbit 36 000 km above the Earth's surface. At this altitude it takes 24 hours for the satellite to orbit the Earth — the satellite therefore appears stationary as viewed from the Earth. This has the advantage that a single satellite can address a fixed geographic region of the Earth's surface. Currently geostationary satellites form an essential part of global telecommunications systems.

Geostationary satellites are ideal for broadcast services as a single satellite is able to address up to $^1/_3$ of the planet's surface. Interestingly, they can also be used for two-way services direct to individual customers; however, there are two key drawbacks — limited overall system capacity and the need for a large (typically 1.8 m) dish to provide a return channel from the customer's premises. For these traditional systems cost is also prohibitive for all but the most specialist two-way applications, with electronics currently costing about £3500 per customer.

New satellite systems

In a drive to increase overall system capacity and reduce the size (and cost) of customer terminal equipment, the US military and the space system manufacturing community have jointly been developing a new geostationary satellite system which operates at higher frequencies, typically around 20—30 GHz. This new breed of geostationary satellite system should be capable of delivering two-way services direct to customers at data rates between 16 kbit/s and 25 Mbit/s.

However, all geostationary satellite systems suffer from the propagation delay caused by the time it takes for the radio signals to travel up to the satellite and back again. To

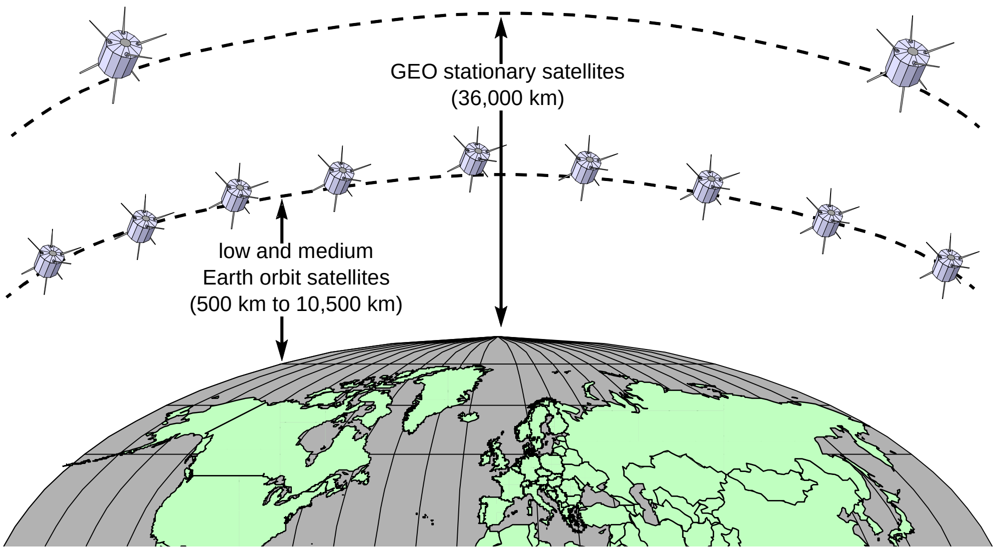


Fig 1 Geostationary satellites orbit the earth at 36 000 km while the emerging breed of low and medium earth orbiting satellites (LEOs and MEOs) orbit the Earth at heights ranging from 500 km to 10 500 km.

overcome these problems, a whole range of medium- and low-Earth orbit satellite systems (MEOs and LEOs respectively) are being developed. A low-Earth orbit is considered to be anything from 500 to 1500 km, while a medium-Earth orbit is about 10 000—10 500 km (Fig 1).

The benefit of these systems is that they do not suffer from propagation delay problems and they offer far greater capacity. They also have the advantage that they can address the mobile handset market whereas the geostationary systems struggle to do this economically.

However, reducing the height of the orbit does produce some difficulties. Firstly, more satellites are needed to cover the Earth. A typical MEO system may require 10 satellites for global coverage while an LEO system might need anything from 50 to 60 satellites or more. Secondly, because these satellites are no longer geostationary, a single satellite cannot address a fixed region of the Earth and this means that all the satellites must be in place before service can begin and, just like any dynamic cellular radio system, some form of hand-off scheme is required to transfer coverage between satellites. The need to launch a full constellation of satellites before any service commences means that LEOs and MEOs require large up-front investments before revenues can be derived. A final technical complexity arises from the need to provide some form of dynamic hand-over between satellites.

Despite these problems various consortia are actively developing systems aimed at providing both mobile narrowband voice and data services and fixed broadband data and multimedia services.

For example, in the narrowband arena, Globalstar, led by Loral Space Systems and Alcatel, plans to provide narrowband voice and data services across the planet. Similarly ICO, a spin-off from Inmarsat, which has about 50 investors including BT, plans to provide global voice and data services. The most advanced of all systems to date is known as Iridium. Iridium is led by Motorola, and now has 66 satellites in a low-Earth orbit, and as of 1 November 1998 will provide narrowband voice and data services to mobile handsets around the globe.

In the broadband world, the Teledesic and Skybridge consortia are considering constellations of many tens of satellites (Teledesic plan to launch 288 satellites in a low-Earth orbit!) aimed at providing two-way broadband data and voice services direct to customer sites at data rates up to 64 Mbit/s.

In total, the various consortia driving these GEO, LEO and MEO initiatives have announced plans to invest upward of $20bn in the next three to five years. A key question for existing fixed network and cellular radio operators is: ‘How will these new networks have an impact upon existing markets?’

If all these planned systems are successful, in a three to five year time-frame we can anticipate a total narrowband global capacity of around 300 000 voice/data channels, of which about 1000 could be addressing the UK. For comparison, this level of capacity represents around 1% of the current UK cellular network capacity. In the broadband arena we anticipate a global capacity of around 200 000 × 2 Mbit/s equivalent circuits of which about 2000 will be available in the UK. This represents roughly 3% of BT’s total current private circuit capacity (some 80 000 × 2 Mbit/s equivalent circuits).

In summary, emerging LEOs/MEOs/GEOs should not be viewed as mass alternatives to a fixed network infrastructure. Rather, they represent an important and extremely flexible means of delivering narrowband or broadband capacity anywhere on the planet. They will find application as an alternative access means for remote areas and as an extension (perhaps temporary) to fixed network infrastructure.

High-altitude platforms

There has recently been much media speculation about the concept of high-altitude platforms (HAPs). These are aircraft or balloons which hover over a city, region or country to provide a telecommunications switch (Fig 2).

There are various organisations driving this concept and various different approaches being explored. At present one of the most interesting propositions is that from Angel Technologies Corporation which is developing a manned aircraft to fly in circles above a city, with an on-board switch using a 28 GHz point-to-multipoint wireless channel to provide coverage over a region of about 75 miles. It is claimed that by the year 2000 one aircraft will be able to provide around 16 Gbit/s (or 4000 × duplex 2 Mbit/s equivalent circuits) capacity to customers in a 65 to 120 km diameter area. Each customer could receive service at data rates up to 25 Mbit/s.

Other consortia are discussing the use of balloons and even a hybrid balloon/helicopter. Fantastic as they sound, none of these ideas violate the laws of physics and thus cannot be ignored as potentially viable means of providing high capacity services in metropolitan regions.

2.2 *Local multipoint distribution system*

Local multipoint distribution system (LMDS) is a broadband wireless access technology capable of delivering 155 Mbit/s to, and 55 Mbit/s from, a customer within a 2 km radius of an LMDS transmitter [1]. Interest in this technology is being driven by the US government's recent auction of approximately 1 GHz of radio spectrum. In the USA competitive local area carriers are viewing this technology, and spectrum, as a means to attack incumbent Regional Bell Operating Companies (RBOCs) in some of their key markets, particularly the small and medium business market.

LMDS is a multipoint radio system, which means that the system's total capacity (of the order of 1000 Mbit/s) can be shared between many users within a 2 km radius of the transmitter. If customer densities are high enough, this 'transmitter sharing' translates into a significant cost advantage as compared to many other access technologies. For instance, analysis indicates that if more than 15 customers in a cell require more than 1 Mbit/s of access, then LMDS compares favourably with current point-to-point radio access technologies (Fig 3). At higher customer densities LMDS becomes an even more attractive option.

There are a number of credible suppliers of LMDS today and the activity in the USA makes it likely that we will see rapid deployment of this technology in key US business markets. As the LMDS market grows we can expect system costs to fall and thus it is likely to become an even more attractive access option.

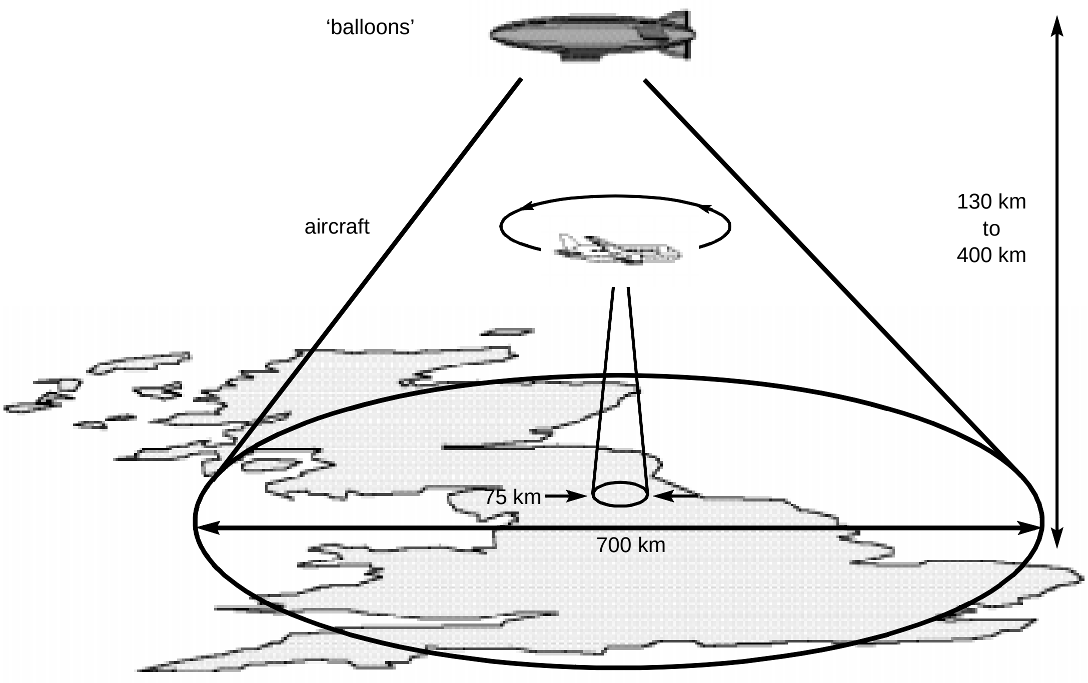


Fig 2 The high altitude platform (HAP) concept uses an aircraft, balloon or hybrid balloon/helicopter in order to hover over a city, region or country to provide a telecommunications switch. If successful, these systems could provide significant communications capacity over a limited geographical area.

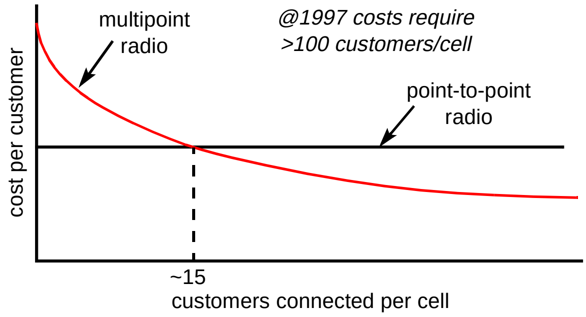


Fig 3 An illustration of the variation of 'cost per-customer-connected' for both a point-to-point radio access system and a point-to-multipoint radio system such as LMDS. It should be noted that the per-customer cost for a point-to-point solution remains constant, while in the multipoint case costs vary as a function of the number of customers connected. Assuming 1997 equipment costs, the multipoint solution becomes cheaper if more than ~15 customers require service at >1 Mbit/s in each of the transmitter areas (~ 2 km radius cell). Applying usual service penetration and radio coverage criteria, this would mean each cell must contain something like 100 such customers.

In the UK the main barrier to the deployment of LMDS is the lack of radio spectrum. However, spectrum does exist which could be licensed for exploitation by this technology should the industry push in this direction. The current lack of spectrum means that LMDS cannot be used in the UK in the very near term; however, in the three to five year time-frame, the economics of this technology combined with the US activity mean that LMDS could compete with existing wireline operators. Essentially LMDS could be used to construct wireless-based metropolitan area networks with similar capacity to first generation metropolitan fibre networks, of the type built by recent new entrants, but requiring only a fraction of the capital expenditure required to install new fibre and duct. However, in the longer term LMDS will not be able to provide the capability of the next generation of higher capacity WDM-based fibre systems.

2.3 Cable modems

Cable modem systems allow cable TV network operators to deliver data services over CATV networks. Currently there are several proprietary systems offering users who are connected to CATV networks a share of bandwidths up to 30 Mbit/s downstream (i.e. from network to user) and 500 kbit/s or so upstream (user to network direction). For example, Motorola have around 1 million modems installed world-wide (mostly in the USA), which are generally used to provide high-speed residential Internet access.

However, very large-scale deployment of proprietary modems has been severely hampered by the lack of interoperability between the various vendors' systems. In 1996, led by the four largest CATV operators in the USA, the cable TV industry formed a group aimed at overcoming many of these vendor interoperability problems.

The Multimedia Cable Network Systems (MCNS) initiative was created with the goal of pushing for fast data access over cable networks. The group has established a series of cable modem design specifications, known as DOCSIS (Data Over Cable System Interface Specification), against which vendors can build.

However, DOCSIS is not just about high-speed Internet access, it is also addressing the real-time services market through its PacketCable™ initiative which aims to provide Internet protocol based voice services over the DOCSIS systems.

More importantly, DOCSIS is also addressing those CATV customers who do not have a personal computer (PC). The group recognises that while nearly everyone has a television there are still huge numbers of people who do not have access to a PC. DOCSIS enables people to access the World Wide Web initially through their televisions, and regards the PC as a 'bolt-on extra' for the enthusiast. Effectively, they are taking two devices — a cable modem and a CATV set-top box — and combining them into one. All major vendors are planning to launch 'DOCSIS-compliant' products in 1998.

From a purely technical standpoint the new range of DOCSIS cable modems offer impressive capability. These systems will support a total downstream capacity of around 240 Mbit/s from a single CATV hub, which in a typical CATV architecture would be shared between 2500 homes. The upstream capacity would be around 40 Mbit/s. For instance, if an existing UK cable operator were to deploy this technology across a typical franchise area, they could, in principle, offer ~100 kbit/s of capacity to each customer passed, or, assuming a 20% take-up, an impressive 0.5 Mbit/s to each customer connected.

In the standards arena there is an interesting complication that has the potential to slow down the deployment of cable modems in Europe. Recently, a group of European cable companies has begun promoting a rival known as the DVB/DAVIC standard. Currently its technical capabilities are similar to DOCSIS, but the equipment is at a much earlier stage of development. DVB/DAVIC-compliant equipment is not expected to be commercially deployable until the latter part of 1999. It is not yet clear which of the two competing cable modem standards will be adopted in the UK and continental Europe, though DOCSIS appears to be more favoured at present.

However, the key question for the CATV industry relates more to the economics of deployment rather than simple technical capability. To deploy any form of cable modem technology, CATV operators would have to 'condition' their networks, requiring significant up-front capital investment. It seems likely that there will be regions in which this investment can clearly be justified — the question for the CATV industry is: ‘On what scale will DOCSIS or DVB/DAVIC cable modem deployment be viable?’ This is a question with no clear answer as yet.

2.4 Splitterless asymmetrical digital subscriber line

A standard telephony copper access system comprises a local exchange, a line card in the exchange, and a telephone wire running from exchange to the customer’s house. These standard ‘copper pairs’, of which BT has more than 40 million, currently underpin BT’s telephony business, delivering analogue voice services into some 26 million homes around the UK. Similarly, elsewhere in the world, it is these copper access lines that form the bedrock of global telephone and data networks. ADSL (asymmetrical digital subscriber line) is a technology which has been developed over the past ten years to overlay a high-capacity data channel on top of the standard analogue voice channel [2]. Typically ADSL offers a 2 Mbit/s data channel to the customer and a return channel of around 300 kbit/s in addition to the standard voice circuit (Fig 4). It provides a cost-effective way to enhance the capability of one of BT’s (and any other telco’s) major assets — its copper local loop.

A key element of standard ADSL is a ‘splitter’ located at the customer premises which provides isolation between the high-speed data channel and the standard voice channel. Many telcos are currently trialling this technology, including BT which plans to run a customer trial in north London. In recent months there has been a blaze of publicity heralding a variation of ADSL which is seen as fundamentally altering the economics of this technology. This variation is known as ‘Splitterless ADSL’, ‘Universal ADSL’ or ‘DSL Lite’, and, as the name suggests, removes the ‘splitter’ from the customer premises equipment.

The basic idea is to trade performance for simplicity and cost. The argument suggests that removing the customer splitter will impair performance slightly (downstream capacity of 1 Mbit/s and upstream of 100 kbit/s), but will dramatically reduce the installation costs through eliminating the need for an engineer to visit the customer premises.

Key industry players, such as Microsoft, Intel, and many modem manufacturers, are championing DSL Lite, as they see it as a direct extension of the existing voice-modem business model. In this model a customer will purchase a modem from a high street retail outlet, install it in their PC and then simply plug it into the telephone socket. In principle, DSL Lite would allow a similar process but would allow Internet access at rates up to 1 Mbit/s rather than today’s 56 kbit/s. At no point does the customer have to arrange with their telecommunications supplier for an engineer to visit and install any equipment (though some conditioning work would be needed at the local exchange).

However, recent measurements conducted at BT indicate that the removal of the customer splitter can lead to degraded and somewhat unpredictable performance. Both the data channel and the analogue voice channel can be impaired in a way which is dependent on the type of

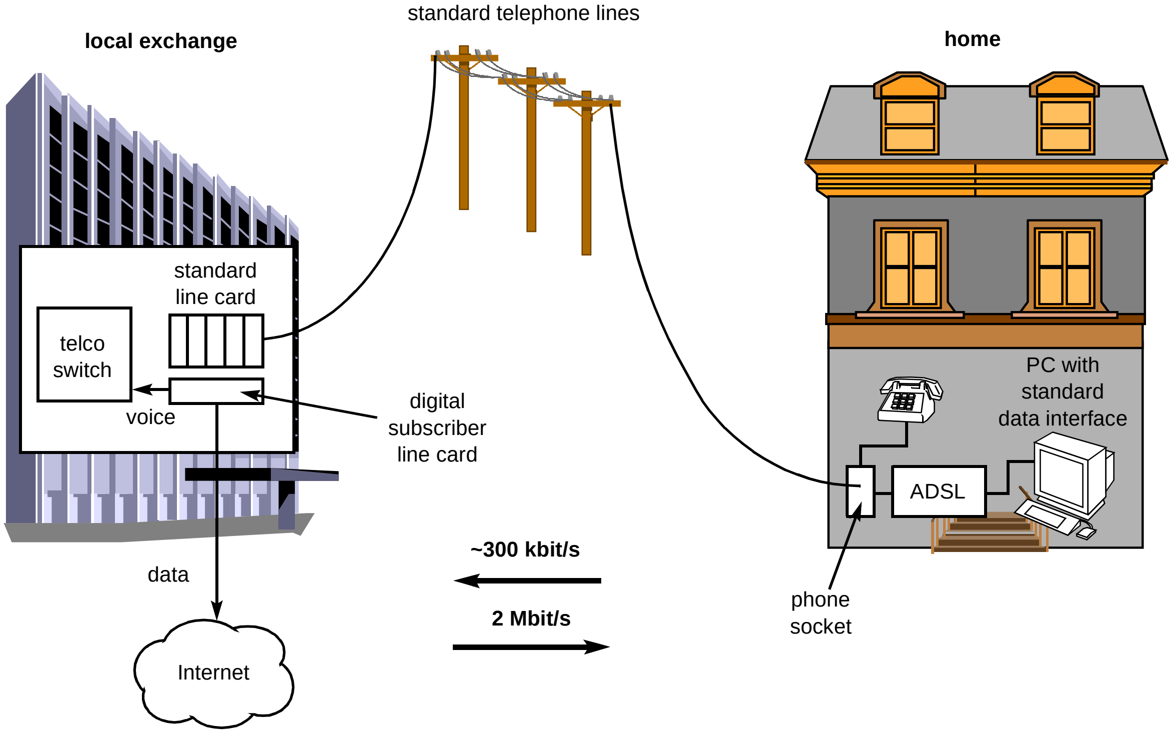


Fig 4 ADSL enhances the capability of a standard copper access line by overlaying a high-capacity digital data channel on top of the standard analogue telephony channel.

telephone, number of telephones, type of home wiring, and even possibly the particular batch of telephones installed. Hence, while the telephone operator saves the capital cost of the DSL Lite modem and the operational cost of a customer installation visit, other operational costs such as those associated with help-desks and fault finding will increase.

At present many proponents of DSL Lite are focusing on the use of ‘micro-filters’ or ‘in-line filters’ to alleviate the performance problems that can be encountered in a purely splitterless configuration. This involves the customer plugging in a very simple, small and cheap low-pass filter into every phone socket in the home. This provides the necessary isolation between the standard telephony channel and the DSL data channel while still maintaining the ability for the customer to self-install the DSL Lite modem. This appears to be an attractive compromise between the performance problems of DSL Lite and the installation costs of ‘normal’ ADSL. Initial trials of DSL Lite are planned to start in the USA in October 1998. Telephone companies are now focusing on quantifying the real-world performance and operational issues, and therefore costs, associated with this technology. In addition, issues such as the appropriate approvals processes, sales distribution channels and consequences of terminating the network in the customer’s PC (i.e. implications for fault-finding, damage liability, workforce IT skills) are being examined.

2.5 Internet through the electricity supply

In October 1997, Nortel, in collaboration with Norweb, announced a new data networking technology called PowerLine. The technology is aimed at the delivery of high-speed data services over existing low-voltage electricity distribution networks. Operation at rates up to 1 Mbit/s (eight times ISDN transmission rates) positions the technology as an alternative for Internet delivery.

In assessing the potential which this technology has to play in future communications networks, it is important to recognise the impact which the significant architectural differences between a telephone network and an electricity distribution network could have. In a telephone network a dedicated line goes from each customer to the local exchange, thus the entire capacity of that line is available to each customer. In contrast, in a mains electricity distribution system power is fed from a local sub-station down a shared tree-and-branch cable network. This means that PowerLine’s 1 Mbit/s capacity must be shared between the 50 to 200 homes connected to a typical distribution system. Thus the actual capacity delivered to each customer is likely to be significantly less than the 1 Mbit/s peak capacity.

In common with any new broadband access technology there are significant economic barriers to the large-scale deployment of this technology. A clear advantage of this approach is the ability to reuse an existing infrastructure of ducts and wiring right into the customer premises. However, engineering works are still required in the customer’s premises in order to install a conditioning unit, which is needed in order to separate the high-speed data channel from the low frequency 240 V mains. It is also likely that major network build would be required to connect the local electricity sub-stations (which may be within 50 m of the customer’s house) back to an Internet point of presence or another trunk network. In addition it is worth noting that, for this technology, the system must cope with significant amounts of impulsive noise, and be engineered to limit radio emissions to acceptable levels [3].

Finally, although protocols, such as the Internet’s TCP/IP, could tolerate the hostile nature of a mains electricity network the delivery of real-time services such as voice or video (or even IP voice) would be significantly degraded by impulsive noise. Nevertheless, this is an interesting technology that, by reusing the existing electricity distribution infrastructure, could provide a lower cost option for a new entrant into an established telecommunications market-place.

3. Technology for the future

3.1 Fibre and radio

Fibre to the home has been inhibited by the cost of opto-electronics and civil engineering costs. Though the cost of equipment may eventually be solved by the economics of scale, the civil engineering cost will still be a rate limiter, particularly for the last few hundred metres from the cabinet to the home. Radio access direct from the central office would avoid a large part of the civil engineering, but spectrum would be too limited for high-density broadband application. An attractive hybrid is to use fibre to connect directly to a radio base-station at either the cabinet site or distribution point. Furthermore, this strategy could provide an interesting synergy with cellular mobile communications and help support forecast growth in mobile multimedia and data applications. Increased capacity from the cellular network leads to greater spectral reuse, more base-stations and hence micro- and pico-cellular systems. It could be possible to use for a common fibre access infrastructure to support both a fixed radio access system and a cellular roaming system. From a technical viewpoint it is also interesting that fibre can be used directly to transport radio and microwave signals in analogue format antenna remoting.

The marriage between fibre and radio is at its closest in the passive picocell concept [4]. This is a simple broadband format-independent radio/fibre interface. Direct translation from optical to RF and from RF to optical is achieved without any electrically powered equipment at the antenna site. A schematic of the configuration is shown in Fig 5. The key device is a multiple quantum-well (MQW) electro-absorption modulator, which has the interesting property of being able to both detect and modulate light simultaneously.

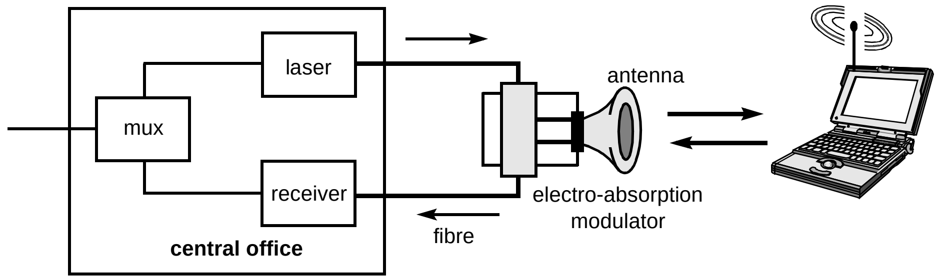


Fig 5 Passive picocell for wireless access — fibre transmission from central office to cabinet or distribution point is transparent to signal format. No electrical power feed is required to the radio base-station.

Because there is no need for an electrical power supply at the remote transducer, any metallic connection between transmitter and receiver is avoided. The latter could have additional advantages in hazardous environments and situations where it is necessary to avoid conduction (high voltage power lines and EMC). Experiments have shown that Mbit/s radio transmission over a line-of-sight distance of about 100 metres is possible with practical optical sources.

The passive picocell offers transparency to signal format, which could allow smooth upgrading without physical disturbance to the network. it also extends the access network into the building and could challenge the need for extensive customer premises wiring in the future.

3.2 Local area network (LAN) technology

A further degree of blurring between access networks and customer premises equipment will come from advances in LAN technology. Standard Ethernet is very widely used in business and is starting to find its way into the home. In large organisations 100 Mbit/s Ethernet is now common place and Gigabit Ethernet is starting to become available [5]. Campus data networks are starting to carry telephony as well as computer traffic. LAN technology is starting to find its way into the wide area. It is only a matter of time before LAN technology is seen carrying out roles normally expected of the access network. It has also been shown that Gigabit Ethernet can operate over the wide area — distances greater than 100 km have been achieved in laboratory trials. In the longer term, LANs are expected to run at even faster speed. A recent experiment at BT Laboratories linked workstations at 2.5 Gbit/s rate to a 40 Gbit/s optical backbone [6]; this is by no means the final limit. Within five years, laboratory demonstrations of terabit computer networks could be anticipated.

3.3 Fibre to the home, WDM and the switchless network

One of the most dramatic changes to core transmission in the past two years has been the development and deployment of wave division multiplexing (WDM) to increase capacity on already installed fibre cables. WDM technology for the core is still an expensive, low-volume technology. However, there is now seen to be a recognised need, particularly in the USA, for WDM within metropolitan networks — this trend will increase volume, thus lowering costs, and extend the penetration of fibre to business customers. Ultimately WDM in the access network could allow huge capacities over comparatively low fibre-count cables. This would not only lead to a fully future-proof network — even able to meet the needs of interactive wall screen TV — but it would also reduce the number of physical connections within the access network. Joints are a main cause of unreliability in today’s copper access network. Speculation on the potential operational benefits is not just limited to a reduced cable fault rate; new optical network architectures could be considered that would also reduce the amount of switch hardware and network management. One radical goal, proposed by Hill [7], is the so-called ‘switchless network’. In this scheme the use of passive optical routeing devices in the centre of the network is combined with time division multiple access (TDMA) and wave division multiple access (WDMA) around the edge. It has been predicted that such a network could be used to switch the traffic of 20 million customers — in effect, an access network with nation-wide reach!

4. Conclusions

4.1 Coming soon

Networks in the sky — the next two years will see the emergence of a whole new breed of networks in the sky based on LEOs, MEOs and GEOs. These satellite-based networks do not represent a large-scale alternative to fixed network infrastructure but rather highly valuable capacity which can be used geographically to target customers anywhere on the planet.

High altitude platforms — in the medium term these systems could provide a credible way to provide broadband capacity in metropolitan areas. HAPs could be a particular threat to established network operators if they are combined with emerging satellite systems to provide a network of interconnected broadband islands hovering over major metropolitan areas.

LMDS — the next few years are likely to see a range of companies (mostly in the USA) experimenting with LMDS-based networks as an attack technology to address specific market opportunities. In the longer term this broadband technology should be viewed as a serious competitive threat to incumbent network operators in areas where significant groups of customers (>100) require bandwidths in excess of 1 Mbit/s. LMDS could therefore find widespread application in larger metropolitan areas.

Cable modems — the raw bandwidth of the new breed of DOCSIS cable modems, the US drive for this technology and its large-scale deployment around the world mean that cable modems must be viewed as a significant access technology where cable TV networks have been deployed.

Splitterless ADSL — though questions exist regarding the technical viability of Splitterless ADSL, the industry push for this technology means it is likely that solutions will be found which enable it or a variant to be used in some European networks. This chance to provide high-speed data services over existing telephony-carrying lines (both traditional telco and cable TV based) clearly provides both a huge potential opportunity and a potential threat and is one which all players must assess as a matter of urgency.

PowerLine — though technically viable, using the electrical distribution network is unlikely to deliver a fundamental change to the economics of high-speed Internet delivery in areas which already have an established telecommunications infrastructure. However, PowerLine does represent an interesting alternative method to enter existing markets without large-scale infrastructure purchase or construction and, as such, may well be used in niche markets.

4.2 Wildcards

Gigabit Ethernet — as this product increases in reach, it is likely we will see metropolitan area networks being constructed from this technology.

Passive picocell — novel short distance radio technologies could replace final drop, and home and office wiring. True fixed/mobile convergence may then be possible at the physical layer.

4.3 Finally, the dream lives on

WDM optical technology in the access network, the whole-life economic benefit of fibre at last realised — access would no longer be a network bottle-neck to new and imaginative broadband services which require high-peak symmetrical bandwidth.

References

1 Merrett R P, Beastall P V E and Buttery S J: ‘Wireless local loop’, BT Technol J, 16, No 4, pp 101—111 (October 1998).

2 Foster K T, Young G, Cook J W, Clarke D E A, Kirkby R H and Booth M G: ‘Realising the potential of access networks using DSL’, BT Technol J, 16 No 4, pp 34—47 (October 1998).

3 Rickard R: ‘The co-existence of wideband carriers above 2MHz on the LV distribution network with radio systems’, PowerLine Telecommunications Forum, Dusseldorf, Germany (November, 1997).

4 Wake D and Moodie D G: ‘Passive picocell — a future integrated wireless communications network’, British Telecommunications Eng J, 16, No 2 (July 1997).

5 http://www.gigabit-ethernet.org/

6 Lucek J K, Gunning P, Moodie D G, Smith K and Pitcher D: ‘SynchroLan: a 40 Gbit/s optical-TDMA LAN’, Elect Letts, 33, p887 (1997).

7 Hill A M, Cotter D C, Hawker I and Tandon V: ‘New network infrastructures for the 21st Century: Part 2 — Ultrafast self-routeing structures’, British Telecommunications Eng J, 13, No 3 (1994).

Don Clarke joined BT Laboratories in 1973. Based in Ipswich, he worked on the development of the Pathfinder digital exchange — a forerunner of System-X. He graduated in Computer Systems from the University of Essex in 1981. Until 1995 he led the fibre systems group responsible for the development of narrowband passive optical networks, and holds a number of patents relating to that technology. In 1995 he transferred to the copper access group and is the project manager for the ADSL and VDSL development activities. He has chaired the international FSAN VDSL working group for the past two years. He is a member of the IEE and IEEE.

Since mid-1997, Greg Lampard has worked on the strategic assessment of access network technologies at BT Laboratories. These include fibre, radio, twisted pair and hybrid fibre/coaxial cable systems. His recent focus has been on high-speed data network design for Internet protocol networks, and performance and cost analysis of cable modem systems. Prior to this he worked in the cable networks section, Telstra research laboratories, Australia where he was involved in laboratory testing, field trials and installation of new technologies, particularly fibre and cable TV systems.

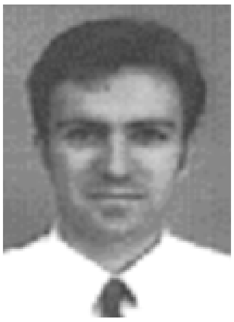
He graduated at the University of Melbourne in 1985 with first class honours in Physics and gained his PhD in Physics there in 1990.

Bob Merrett joined BT as an apprentice in 1965. After graduating from the University of Wales with a degree in Electrical and Electronic Engineering in 1969 and a PhD from the University of Cambridge in 1972, he joined the reliability division at BT Laboratories where he led teams assessing the reliability of semiconductor packaging, reed relays and a range of discrete electronic devices. In 1983 he joined the devices division and led a team designing high speed GaAs integrated circuits and then a team developing InP OEICs. Since 1990 he has worked on various aspects of radio local loop and currently co-ordinates the technical projects on wireless access and local area mobility.

David Smith joined BT Laboratories (BTL) in 1967 as an apprentice. In 1974 he graduated from Brunel University with a degree in Electrical Engineering. He worked on some of the first trials of optical transmission systems in 1978 and then went on to lead a group researching new directions for exploiting single-mode optical communications. The team carried out pioneering work in long-distance communication and went on to develop new concepts for optical processing and networking. Through the late 1980s he led a large long-term research project that carried out seminal work on concepts such as WDM, fibre amplifiers and optical switching. He now manages the Networks Research unit at BTL and has responsibility for activities ranging from device fabrication, physical network layer research through to network protocols. He has authored or co-authored about 100 papers and 20 patents.

Tim Whitley joined BT in 1981 as an apprentice in North Wales. Upon completion of his apprenticeship he obtained sponsorship to study Physics at the University of Kent. Graduating in 1987 he joined the optical physics division at BT Laboratories and during the subsequent eight years conducted research into various aspects of optical telecommunications. In particular he was involved in BT's pionerring research and development into optical fibre amplifiers and their application to future photonic networks. Since 1997 he has worked in the technology assessment unit (part of N&S technology directorate) where he provides technical consultancy to many parts of the business and produces senior management briefing material on a diverse range of technology issues. He has published more than fifty scientific papers and holds a PhD in Electronics Systems Engineering.

Jon Wakeling has worked in the communications industry for twenty years. He joined BT Laboratories in 1989 after graduating from Plymouth Polytechnic with a degree in communications engineering. He has specialised in satellite communications systems with a particular focus on mobile satellite system performance enhancement and Ka-band system technology. He is currently working on the assessment of future broadband satellite systems and related network management requirements. He manages the satellite systems team at BT Laboratories. He is a Chartered Engineer and a Member of the IEE.



Gavin Young received a BSc in Electronic Engineering from the University College of Swansea and an MSc in Communications Engineering from the Imperial College, London University in 1986. He is technical area leader for BT's copper access team. He is also on the Board of Directors of the ADSL Forum and has led the network migration and VDSL working groups within the Forum.

He was recently appointed as overall technical chairman of the ADSL Forum. He is a Member of the IEEE and an associate member of the IEE.